

CHAPTER 2 LITERATURE REVIEW AND RELATED WORK – Guidelines (BECE & BSIT Programmes)

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2.0 Introduction

In computer science and computer engineering research, a literature review is a structured analysis of previously published research related to a specific technical problem. Rather than simply summarizing earlier studies, the literature review critically evaluates existing **algorithms, technologies, system architectures, and methodologies** to understand how researchers have attempted to solve the problem.

The main purpose of this chapter is to:

- demonstrate understanding of existing technologies and research
- analyse previous solutions and their limitations
- identify gaps in current knowledge
- justify the need for the proposed system or solution

In computing disciplines, the literature review must focus on **technical analysis**. Students should examine how different systems work, compare their performance, and evaluate their strengths and weaknesses.

This chapter, therefore, reviews existing research related to the **specific project topic**, examining both traditional and modern approaches used to solve the identified problem. The chapter concludes by identifying a **research gap** and introducing the conceptual direction of the proposed solution.

Important guideline

The introduction must immediately state the **actual project topic**. Avoid generic statements such as “*this chapter reviews literature related to the topic.*”

Example:

Instead of writing:

This chapter reviews literature related to network security.

Write:

This chapter reviews existing intrusion detection systems designed for small-office networks and analyses their effectiveness in detecting cyber threats.

Example of an introduction

This chapter presents a critical review of existing research related to **[insert project topic]**. The purpose of this review is to examine previously proposed solutions, analyse their strengths and weaknesses, and identify gaps that justify the need for the proposed system.

Recent advances in computing technologies have enabled the development of various systems designed to address **[describe the general problem]**. However, despite these advances, several technical challenges remain, particularly in areas such as **[accuracy, scalability, real-time processing, cost efficiency, automation, security, etc.]**.

Researchers have proposed different approaches to address this problem, including **[machine learning techniques, IoT-based systems, cloud architectures, embedded systems, heuristic algorithms, etc.]**. While these approaches demonstrate promising results, they also present limitations related to **[performance, computational cost, reliability, scalability, or adaptability]**.

This chapter reviews and analyses these approaches in order to identify the **research gap** that motivates the present study.

The chapter is organised as follows:

- Section 2.1 presents the theoretical background of the study.
- Section 2.2 reviews existing approaches used to address the problem.
- Section 2.3 provides a comparative analysis of existing systems.
- Section 2.4 identifies the research gap.
- Section 2.5 introduces the conceptual direction of the proposed system.

2.1 Theoretical Background

This section introduces the key concepts and technologies necessary to understand the research problem.

Example:

2.1.1 Definition of Key Concepts

Concept 1: [Example: Machine Learning]

Machine learning refers to computational techniques that allow systems to learn patterns from data and make predictions without explicit programming (Author, Year).

Machine learning algorithms are widely used in applications such as:

- image recognition
- intrusion detection

- predictive analytics
- natural language processing

These algorithms can generally be classified into:

- supervised learning
- unsupervised learning
- reinforcement learning

Concept 2: [Example: Internet of Things]

The Internet of Things (IoT) refers to interconnected devices capable of collecting and exchanging data through the internet (Author, Year).

IoT systems typically consist of:

- sensors
- communication networks
- processing units
- data analytics platforms

IoT technology enables real-time monitoring and automation in various domains such as agriculture, healthcare, and smart cities.

Concept 3: [Insert Other Relevant Concepts]

Students should define **3–5 important technical concepts** related to their project.

2.2 Review of Existing Approaches

This section examines previous research conducted in the problem domain. The review is organised based on the **evolution of technologies and methodologies** used to address the problem.

2.2.1 Traditional Approaches

Early solutions to **[problem]** relied on conventional techniques such as:

- rule-based systems
- statistical analysis
- manual monitoring
- deterministic algorithms

For example, **Author (Year)** proposed a system that used **[method]** to address **[problem]**. The system improved **[specific performance metric]**, but it suffered from limitations such as:

- low scalability
- high computational cost
- lack of adaptability

Similarly, **Author (Year)** developed **[type of system]** designed to improve **[specific function]**. While the system achieved **[specific improvement]**, it lacked **real-time capability or automation**, which limited its practical use.

These traditional systems formed the foundation of modern computing solutions but often struggle to meet the demands of **large-scale and real-time environments**.

Important Rule

Every paragraph in a literature review must include **citations from actual research papers**

Example :

Early research addressing **[problem]** relied primarily on traditional computational techniques.

For example, **Author (Year)** proposed a **[system or algorithm]** designed to **[describe purpose]**. The system demonstrated improvements in **[specific metric]**, but it suffered from limitations such as **[low scalability, limited accuracy, high computational cost, etc.]**.

Similarly, **Author (Year)** developed a **[method/system]** that used **[specific technique]** to address **[problem]**. Although the system improved **[performance metric]**, it lacked **[real-time processing, automation, adaptability, etc.]**.

These traditional methods provided important foundational work; however, their limitations prompted researchers to explore more advanced computational techniques.

Students should discuss **4–6 research papers** in this section.

2.2.2 Intelligent and Data-Driven Systems

Recent research has increasingly focused on **intelligent computing technologies** to improve system performance.

Common technologies include:

- Machine Learning
- Deep Learning
- Internet of Things (IoT)
- Cloud Computing
- Edge Computing
- Big Data Analytics

For example, **Author (Year)** developed a **machine learning-based system** capable of predicting **[event/condition]** using algorithms such as:

- Random Forest
- Neural Networks
- Support Vector Machines

The results showed improved prediction accuracy compared to traditional statistical methods.

Another study by **Author (Year)** proposed an **IoT-based monitoring framework** where sensor data is transmitted to a central processing unit for analysis. This approach enabled:

- real-time data collection
- automated decision making
- remote monitoring

Although intelligent systems provide significant improvements, several challenges remain:

- high computational requirements
- large data requirements
- security vulnerabilities
- system complexity

Example:

Recent studies have increasingly focused on intelligent computing techniques such as machine learning and artificial intelligence to address the limitations of traditional systems.

For instance, **Author (Year)** proposed a machine learning model using **[algorithm name]** to predict **[event or outcome]**. The system achieved **[performance results such as accuracy or processing speed]**, demonstrating improved performance compared to earlier methods.

Another study by **Author (Year)** introduced an **IoT-based monitoring system** where sensor data is collected and transmitted to a central server for analysis. This approach enabled **real-time monitoring and automated decision making**.

Despite these advancements, intelligent systems still face challenges such as:

- high computational requirements
- dependence on large datasets
- security vulnerabilities
- system complexity

Students should analyse **4–6 modern research papers** in this section.

2.2.3 Emerging Technologies

Example:

Recent developments in computing have introduced emerging technologies that further enhance system capabilities.

These technologies include:

- edge computing
- blockchain systems
- federated learning
- cloud-edge hybrid architectures

For example, **Author (Year)** proposed a **cloud-based architecture** designed to improve system scalability and data processing efficiency. While the approach improved **processing speed**, it introduced challenges related to **network latency and security**.

Similarly, **Author (Year)** developed an **edge computing framework** that processes data closer to the data source. This approach reduces latency but requires **additional hardware infrastructure**.

Students should analyse **3–4 emerging technology studies**.

2.3 Comparative Analysis of Existing Systems

Different research approaches often present **trade-offs**.

For example:

- Some systems provide **high accuracy** but require significant computational resources.
- Others offer **low-cost implementation** but provide limited predictive capability.
- Some solutions lack **real-time processing**, which is essential in modern distributed systems.

A **comparison table** is recommended to clearly show differences between existing systems. A comparative analysis helps identify the strengths and weaknesses of existing solutions.

Example Table 2.1

Table 2.1: Comparison of Existing Systems

Author	Method Used	Key Contribution/Strength	Limitations
Author A (2021)	IoT Monitoring System	Real-time monitoring	High energy consumption
Author B (2022)	Machine Learning Model	High prediction accuracy	Requires large dataset
Author C (2023)	Cloud-based Architecture	Improved scalability	Network latency

This comparison highlights the need for improved systems that balance **accuracy, efficiency, scalability, and cost-effectiveness**.

Students should include **1–2 comparison tables** in Chapter 2.

2.4 Research Gap

After reviewing previous research, the next step is to identify **what still remains unsolved**.

This is known as the **research gap**.

Despite the significant progress in systems designed to address **[problem]**, several limitations still exist.

For example:

- Many systems focus primarily on **accuracy**, ignoring **real-time performance**.
- Some solutions require **large datasets** that may not be available in practical environments.
- Other systems rely on **high-performance hardware**, making them unsuitable for low-cost deployment.

Additionally, many existing solutions are designed for **controlled experimental environments**, making them difficult to deploy in **real-world conditions**.

Therefore, there is a need for a system that:

- improves **system performance**
- reduces **computational complexity**
- supports **real-time processing**
- remains **scalable and cost-effective**

This unresolved problem forms the **foundation of the present research**.

Please Note:

Avoid vague statements like:

There is limited research in this area.

Instead, clearly explain **what exactly is missing in existing research**.

Example:

Based on the analysis of existing literature, several limitations remain in current solutions addressing **[problem]**.

Most existing systems focus on **[specific aspect such as prediction accuracy or monitoring capability]**, while limited attention has been given to **[important missing aspect]**.

Additionally, many solutions rely on **high-performance hardware and large datasets**, making them unsuitable for environments with limited computational resources.

Furthermore, several systems lack **real-time processing capabilities**, which are essential for practical deployment in real-world environments.

Therefore, there is a need for a system that:

- improves **system accuracy and efficiency**
- reduces **computational complexity**
- supports **real-time data processing**
- remains **scalable and cost-effective**

Addressing this gap forms the **primary motivation for the present research**.

2.5 Proposed Conceptual Solution

After identifying the research gap, briefly introduce the **proposed solution**.

Based on the limitations identified in the literature, this research proposes the development of **[describe the system or method]**.

The proposed system will integrate technologies such as:

- IoT sensors
- machine learning algorithms
- embedded systems
- cloud computing platforms

The goal is to improve **[system performance metric]**.

Expected advantages of the proposed solution include:

- improved system efficiency
- enhanced data processing capability
- reduced manual intervention
- improved accuracy and reliability

The **detailed system design and implementation methodology** will be presented in **Chapter 3**.

Important

Do not explain technical implementation details in Chapter 2. Those belong in the **Methodology chapter**.

Example

Based on the identified research gap, this project proposes the development of **[describe the proposed system]**.

The proposed system will integrate **[technologies used in the project]** in order to improve **[performance metric]**.

The expected advantages of the proposed system include:

- improved system efficiency
- enhanced automation
- reduced manual intervention
- improved accuracy and reliability

The detailed architecture and implementation of the proposed system will be presented in **Chapter 3**.

2.6 Chapter Summary

Example:

This chapter reviewed existing literature related to **[research topic]**, examining both traditional and modern approaches used to address the identified problem.

The strengths and limitations of these approaches were analysed, revealing several unresolved challenges within the field.

Based on this analysis, a **research gap** was identified, highlighting the need for improved solutions that address **[specific limitations]**.

The next chapter presents the **research methodology and system design** proposed to address these challenges.

Recommended Literature Review Requirements for Students

Minimum expectations:

Requirement	Recommendation
Number of sources	15–25 scholarly papers
Recent sources	At least 60% from 2019–2026
Databases/scholarly sources	IEEE, ACM Digital Library, ScienceDirect, SpringerLink, Google Scholar Avoid Wikipedia, blogs, and random websites Use peer-reviewed research papers only .
Pages	8–10 pages
Tables	At least 1 comparison table